

Change Request (CR-01)

Credly Badge Wallet & Skills Wallet Integration

Amendment Aligned to Sulfri Trainer Portal SRS v1.0 (Personal Brand Edition)

Base SRS: Sulfri Trainer Portal — Personal Brand Edition v1.0

CR ID: CR-01

Type: Functional Enhancement (Core Branding Module)

Priority: High

Target Version: v1.1 (Post-Approval Consolidation)

1. Executive Summary

This Change Request introduces a **Verified Credentials Module** consisting of:

1. Badge Wallet (Credly Embedded)
2. Skills Wallet (Derived from Badge Mapping)

The objective is to extend the existing Personal Branding Portal (Profile, Experience, Join Links, Downloads) by adding a **structured, verifiable digital credential showcase** without altering the original architectural foundation.

This CR strictly aligns with the existing system scope, roles, database philosophy, and admin-controlled content model defined in SRS v1.0.

2. Alignment with Existing SRS v1.0

The original SRS defines: - Public profile page - Experience management - Join class links - Downloadable trainer profile - Admin CRUD control over all content - Next.js + API Routes + PostgreSQL architecture

This CR: - DOES NOT introduce new user roles - DOES NOT change authentication model - DOES NOT alter deployment architecture - EXTENDS existing admin-managed content framework

The Badge Wallet is treated as another **admin-managed public content module**, similar to Experience and Join Links.

3. Scope of Change

3.0 Additional Consideration — Dynamic JSON Workaround (Auto-Sync Option)

In addition to manual embed-based badge management, this CR now includes an OPTIONAL auto-synchronization mechanism using Credly's public JSON endpoint format:

`https://www.credly.com/api/v1/users/{USER_ID}/badges.json`

This allows the system to automatically retrieve badge metadata without requiring an official API key.

This feature is classified as: - Enhancement Layer (Optional Mode) - Not mandatory for MVP - Must remain compatible with existing manual embed workflow

Two operating modes must be supported:

Mode A — Manual Mode (Default) Admin pastes embed code and manages badges manually.

Mode B — Auto-Sync Mode (Optional Toggle) System fetches badge data from Credly public JSON endpoint and creates/updates badge records automatically.

Both modes must coexist without architectural conflict.

3.1 In Scope

Public Pages (New)

- `/badges` — Verified Credentials Wallet
- `/badges/{slug}` — Badge Detail Page
- `/skills` — Skills Wallet (Aggregated View)

Admin Pages (Extension of Existing Backoffice)

- `/admin/badges`
- `/admin/skills`

These follow the same CRUD pattern as: - `/admin/experience` - `/admin/join-links`

4. Functional Requirements (Extension to Section 5 of SRS v1.0)

4.0 Badge Operating Modes

The system must support two badge ingestion strategies:

1. Manual Embed Mode
2. Auto-Sync JSON Mode

Manual mode remains primary to ensure stability if public endpoint structure changes.

(Extension to Section 5 of SRS v1.0)

4.1 Badge Wallet — Admin (Manual Mode)

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Admin must be able to create a badge entry by pasting Credly Embed Code.

System must extract only: - data-share-badge-id - data-iframe-width - data-iframe-height - data-share-badge-host

System must NOT store raw script tags.

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Admin must be able to manually input badge ID if embed code is unavailable.

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Admin must control: - Visibility (Public / Hidden) - Featured flag - Display order

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Admin must assign one or more skills to each badge.

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Admin must preview badge render before publishing.

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Admin must batch import multiple badges via embed code paste.

4.2 Badge Wallet — Admin (Auto-Sync JSON Mode)

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Admin must be able to enable Auto-Sync Mode by providing: - Credly User ID - Sync interval (manual trigger / daily cron)

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System must fetch badge data from:
https://www.credly.com/api/v1/users/{USER_ID}/badges.json

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System must: - Parse badge ID - Extract title - Extract issuer - Extract issue date - Extract badge image URL - Generate verification URL

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System must upsert badges: - New badges are created automatically - Existing badges are updated - Removed badges are marked as archived (not deleted)

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Auto-Sync must not overwrite admin-controlled fields such as: - Featured flag - Display order - Skill assignments - Visibility override

4.3 Badge Wallet — Public

4.2 Badge Wallet — Public

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Public users must be able to view a grid-based Badge Wallet page.

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Wallet must support: - Category filter - Search by title/issuer - Sort by newest/featured

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Each badge must include: - Embedded Credly badge - Issuer - Issue date (optional) - Verify button linking to Credly page

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If embed fails, fallback placeholder must display.

4.4 Skills Wallet — Public

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Skills Wallet page must aggregate skills assigned to badges.

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Each skill must show: - Number of supporting badges - Clickable link to related badges

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Skill ordering must be controlled by admin.

5. Database Amendment (Extension to Section 6)

Existing SRS includes tables for profile, experience, join links, settings.

This CR adds the following tables:

5.1 badges

- id (PK)
- title
- slug (unique)
- issuer
- issue_date (nullable)
- expiry_date (nullable)
- credly_badge_id (required)
- credly_host (default https://www.credly.com)
- iframe_width (default 150)
- iframe_height (default 270)
- verification_url (nullable)
- featured (boolean)
- visibility (PUBLIC | HIDDEN)
- display_order (int)
- fallback_image_url (nullable)

- created_at
- updated_at

5.2 skills

- id (PK)
- name (unique)
- visibility (PUBLIC | HIDDEN)
- display_order
- created_at
- updated_at

5.3 badge_skills

- badge_id (FK)
- skill_id (FK)
- Composite unique key (badge_id, skill_id)

These tables follow the same naming and audit pattern used in the original SRS.

6. API Amendments (Extension to Section 7)

6.0 Auto-Sync Service Layer

System must include an internal service responsible for: - Fetching JSON from Credly endpoint - Validating response structure - Handling rate limits - Logging sync activity

Sync must be triggered by: - Manual admin button OR - Scheduled background job (cron)

Error handling must include: - Graceful failure if endpoint unavailable - No crash to public badge pages

(Extension to Section 7)

Public Endpoints

- GET /api/badges
- GET /api/badges/{slug}
- GET /api/skills
- GET /api/skills/{slug}

Admin Endpoints

- POST /api/admin/badges
- PATCH /api/admin/badges/{id}
- DELETE /api/admin/badges/{id}
- POST /api/admin/badges/batch-import
- CRUD /api/admin/skills

All endpoints follow the same authentication model defined in the base SRS.

7. Security Amendment (Extension to Section 8.1)

Additional controls required for JSON Auto-Sync:

1. Validate USER_ID format before saving.
2. Only allow HTTPS calls to www.credly.com domain.
3. Implement request timeout and retry policy.
4. Store last_sync_at timestamp.
5. Log sync failures for admin visibility.

Manual embed security controls remain mandatory.

(Extension to Section 8.1)

Because this module uses third-party embed scripts, additional controls are mandatory:

1. Do not store raw
from admin input.
2. Allow only:
 - https://www.credly.com
 - https://cdn.credly.com
3. Update CSP: script-src 'self' https://cdn.credly.com frame-src https://www.credly.com
4. Validate badge ID format before saving.

These controls integrate into the existing Security Requirements section.

8. Performance Amendment (Extension to Section 8.2)

Manual Mode: - Load embed.js once per badge route. - Lazy load badge components.

Auto-Sync Mode: - Cache JSON response server-side. - Do not call Credly endpoint on every public page request. - Sync must be background-driven, not user-triggered on public page load.

(Extension to Section 8.2)

1. Load Credly embed.js only once per badge route.
 2. Implement lazy loading for badge grid.
 3. Support pagination if badge count exceeds 30.
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9. Acceptance Criteria Amendment (Extension to Section 10)

The feature is considered complete when:

1. Admin can add and publish a Credly badge.
 2. Public Badge Wallet renders multiple badges correctly.
 3. Skills Wallet aggregates assigned skills.
 4. Verification links open official Credly page.
 5. Malicious embed scripts are rejected.
 6. No performance degradation with 20+ badges.
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10. Impact on Deployment

No infrastructure change required. No additional services required. No change to hosting model.

Database migration required for 3 new tables. Frontend route additions required.

11. Version Control Recommendation

Upon approval of CR-01: - Update main SRS version to v1.1 - Merge this amendment into Sections 5, 6, 7, 8, and 10 - Maintain CR log in Appendix

12. Strategic Value Statement

With the addition of Auto-Sync JSON Mode, the portal now supports:

- Fully automated badge updates when new credentials are earned
- Reduced admin workload
- Real-time credibility updates
- Hybrid control (automation + manual override)

This ensures the system remains future-ready while preserving stability and security.

This module upgrades the portal from a static trainer profile to a **verified capability showcase platform**, strengthening:

- Corporate trust
- Procurement credibility
- Government proposal positioning
- Digital professionalism

It remains aligned with the original architectural vision and admin-driven content philosophy defined in SRS v1.0.